CE 310

Monte Carlo Simulation

Week 13 · Session 1 · Propagating Uncertainty Through a Model

University of Arizona · Dr. Wooyoung Jung · Fall 2026

The Schedule Has Ten Days of Float

60 seconds · pick A, B, or C

A plant expansion starts up only when **both** work paths finish. Contract date: **day 190**. Liquidated damages: **$12,000/day**.

Work path	Average duration	Typical spread
Process equipment	180 days	± 12 days
Electrical and controls	176 days	± 14 days

CPM reports a 180-day critical path — ten days of float. **What is the chance you finish late?**

- **A)** Near zero — both paths average well under 190
- **B)** Roughly 15% — the float absorbs most of the risk
- **C)** About one in three

✅ Answer: C — One Chance in Three

The average of the later path is not the later of the average paths

	Deterministic CPM	Monte Carlo, 10,000 runs
Expected startup	day 180	**day 185.5**
90th percentile startup	*not available*	**day 199**
Probability of passing day 190	*not available*	**33%**

Startup waits on whichever path runs late: a bad draw on **either** path delays the plant, while a good draw on one buys nothing. That asymmetry ate 5.5 of the 10 float days before any activity ran long.

> ⚠️ At the 90th percentile, startup is 9 days late — **$108,000 in damages** on a schedule reported as having float.

The Opener Has a Name — Merge Bias

Two paths, each averaging comfortably under the deadline, and the **pair** still ran late. That is not a quirk of those particular numbers. It happens whenever a milestone waits on several things at once.

Startup happens at the **later** of the two finishes:

$$\text{startup} = \max(\text{process},\ \text{electrical})$$

And the average of a maximum is not the maximum of the averages:

$$E[\max(A, B)] \ \geq\ \max\big(E[A],\ E[B]\big)$$

A bad draw on **either** path pushes the date out. A good draw on one buys nothing, because the other path is still running. The penalties accumulate; the credits do not.

> 💡 Every parallel merge in a schedule leaks float this way, and CPM cannot see it — CPM runs on single fixed durations, and a single duration has no bad draw. Ten paths merging is far worse than two.

From One Scenario to Many

Week 12 moved one input at a time and reported a swing. **The new question:** every input is uncertain at once, and uncertain by different amounts. What is the full range of plausible costs — and how likely is each part of it?

Week 12 (OAT)

- One input, one shift
- Answer: a single swing in dollars
- Cannot state a probability
- "Unit price: $59,144 of swing"

Week 13 (Monte Carlo)

- All four inputs random at once
- Answer: a full distribution
- Gives probabilities and percentiles
- "30% contingency holds overrun risk under 5%"

The Monte Carlo Method

Three steps:

1. **Model the input uncertainty:** assign each uncertain input a probability distribution (e.g., take-off quantity ~ Normal(2,000 CY, $\sigma = 182$))
2. **Sample:** draw N random values for each input ($N = 10{,}000$ is typical)
3. **Run the model for each sample:** produce N output values, forming an empirical distribution of outcomes

For a cost model this is fast — just array arithmetic. For a model that takes minutes per run, such as a finite-element analysis or a discrete-event schedule simulation, $N = 10{,}000$ would take months. Engineers fit a fast surrogate first, and the regression from Week 11 is exactly that.

Choosing a Distribution

Monte Carlo needs a **shape**, not just a range — and the shape is a modelling choice you have to defend.

Distribution	Use it when	Typical engineering use
Normal	small errors pile up symmetrically about a centre	measurement error, take-off quantity
Log-normal	the quantity is positive and varies multiplicatively	prices, permeability, fatigue life
Triangular	all you have is a min, a most-likely and a max	expert-elicited durations and costs
Uniform	you know the bounds and nothing about the inside	a crude first pass; rarely the truth

Triangular is the workhorse of construction risk practice — three numbers a superintendent can actually give you over the phone.

> ⚠️ The distribution is an assumption about an input, and it is the one nobody audits. Uniform in particular claims every value in the range is equally likely — almost never what a person means when they hand you a range.

Which Shape Would You Use?

90 seconds · match each input to a distribution

Four inputs on the same job. None of them is Normal just because Normal is the default.

	Input	What you actually know
A	Concrete delivery time	super says "45 min best, 60 typical, 3 hours if the plant backs up"
B	Steel unit price	positive, quoted to you as a percentage swing either way
C	Survey take-off error	symmetric, plus or minus, many small independent causes
D	Depth to bedrock	the borings bracket it; nothing favours the middle

Match each to Normal, log-normal, triangular, or uniform.

✅ A → Triangular · B → Log-normal · C → Normal · D → Uniform

Input	Shape	Why
A Delivery time	**Triangular**	a three-point estimate, and badly skewed right
B Steel price	**Log-normal**	positive, and a "±20%" quote is multiplicative
C Take-off error	**Normal**	many small independent errors — the CLT's home ground
D Depth to bedrock	**Uniform**	bounded, with no reason to prefer the centre

> 💡 A is the one worth arguing about. "45 best, 60 typical, 180 worst" has a **long right tail** — its mean sits near **95 minutes**, well above the 60 anybody would have called typical. Centre a Normal on 60 and you have hidden the delay you were trying to price.

One Draw, Start to Finish

"Sample all four inputs and run the model" is a short sentence hiding a very small act. Here is **draw number one**, in full:

Input	Its distribution	This draw
Quantity	Normal(2,000, σ = 182)	**2,043 CY**
Unit price	log-normal about \$21.26	**\$22.14/CY**
Overhead	Normal(0.13, σ = 0.0304)	**0.145**
Escalation	Normal(0.05, σ = 0.0182)	**0.038**

Push them through the same cost model as last week:

$$2{,}043 \times \$22.14 = \$45{,}232 \rightarrow \times\ 1.145 = \$51{,}791 \rightarrow \times\ 1.038 = \mathbf{\$53{,}759}$$

That is one simulated job. Draw two lands at \$42,574, draw three somewhere else again. Ten thousand of them make a distribution.

> ⚠️ Illustrative draws — your seeded notebook produces its own four numbers. Copy the *procedure*, not these values.

Quick Check — Which Draw Costs More?

60 seconds · reason it out before you multiply

Two draws from the same four distributions:

	Draw A	Draw B
Quantity	2,180 CY	1,890 CY
Unit price	$19.40/CY	$23.80/CY
Overhead	0.11	0.15
Escalation	0.04	0.06

Draw A is **more work at a lower price**. Draw B is **less work at a higher price**.

Which job costs more — and roughly by how much?

✅ B, by About $6,000

The inputs did not cancel — they compounded

Step	Draw A	Draw B
Q × P	$42,292	$44,982
× (1 + OH)	$46,944	$51,729
× (1 + ESC)	**$48,822**	**$54,833**

B costs **$6,011 more** — about 12%.

Watch what did the work. A had 15% more quantity, and B's higher price wiped that out on the first line alone. Then B's overhead *and* escalation both ran high, widening the gap twice more. **All four moved at once, and all four moved the same way.**

> 💡 Precisely what one-at-a-time could not show you. Week 12 moved one input and pinned three; here all four moved together, and the answer depended on how they combined — the only reason a simulation is worth running.

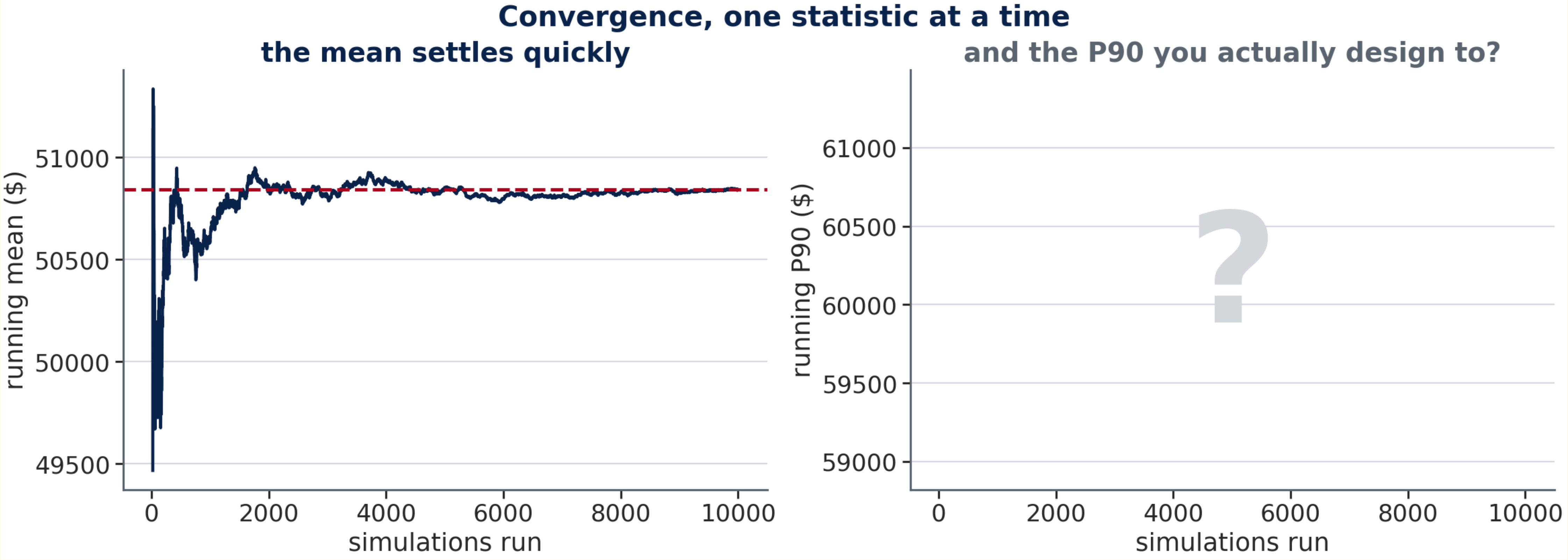
Why N = 10,000?

⚠️ Why N = 10,000 and not N = 100? Tail percentiles (P5, P95) are estimated from only the most extreme samples. At N = 100, P5 is based on roughly 5 data points — noisy and highly seed-dependent. At N = 10,000, P5 is based on roughly 500 points — far more stable. The mean converges fast with small N; the tails need many more samples to settle down.

Monte Carlo is not a special algorithm — it is just systematic sampling from uncertainty distributions. The "simulation" is the repeated evaluation of a model under random conditions.

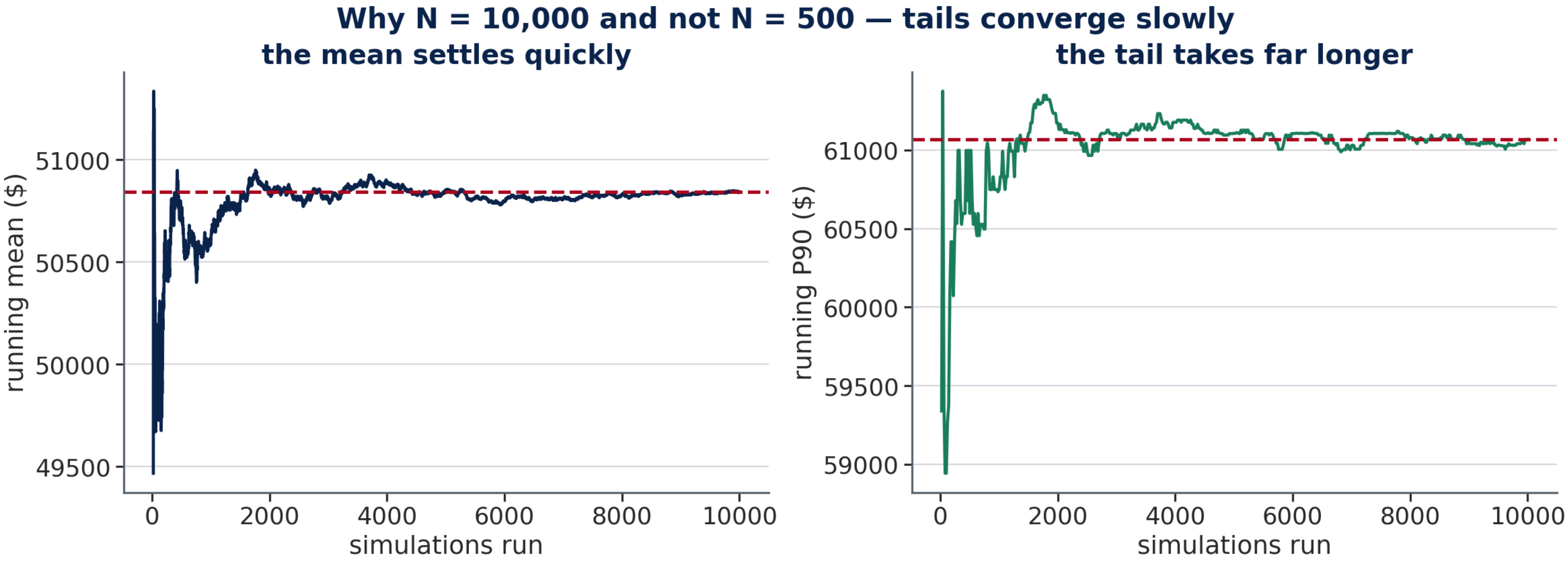
How Many Runs Is Enough?

45 seconds · commit to a number



The mean has settled by a few hundred runs. **But you budget to the P90. How many runs does *that* take?**

Why 10,000 and Not 500



The mean stabilises after a few hundred runs. The P90 is still wandering at several thousand. Whatever you need from the tail, you need more runs to get it.

For Linear Models: Analytical vs. Simulated

For a linear model $Y = a + b \cdot X$ with $X \sim \text{Normal}(\mu, \sigma)$, the output spread is exact:

$\sigma_Y = |b| \times \sigma_X$ — no simulation needed.

Quantity example: a ±15% take-off range, read as a 90% interval, is $\sigma_Q = 182$ CY.

- Analytical: $\sigma_{cost} = 182 \times \$21.26 \times 1.13 \times 1.05 =$ **\$4,601**
- Monte Carlo ($N = 10{,}000$): **\$4,617**

Monte Carlo becomes necessary when the model is non-linear in the input, or the input is not Normal. **Unit price is both** — it comes from a $\log_{10}$ fit.

The Output Distribution — What It Looks Like

Four uncertain inputs, 10,000 simulated jobs, one distribution:

Base cost (deterministic)	$50,459
Simulated **mean**	**$50,844**
Simulated **median**	$50,350
Standard deviation	$7,716
P5 · P95	$38,969 · $64,220

⚠️ The mean sits **above** the median, and above the base cost. That is not sampling noise. Price is drawn log-normally, and a log-normal's mean exceeds its median — a few simulated jobs price very high and pull the average up. The *typical* job did not get more expensive. The *average* one did.

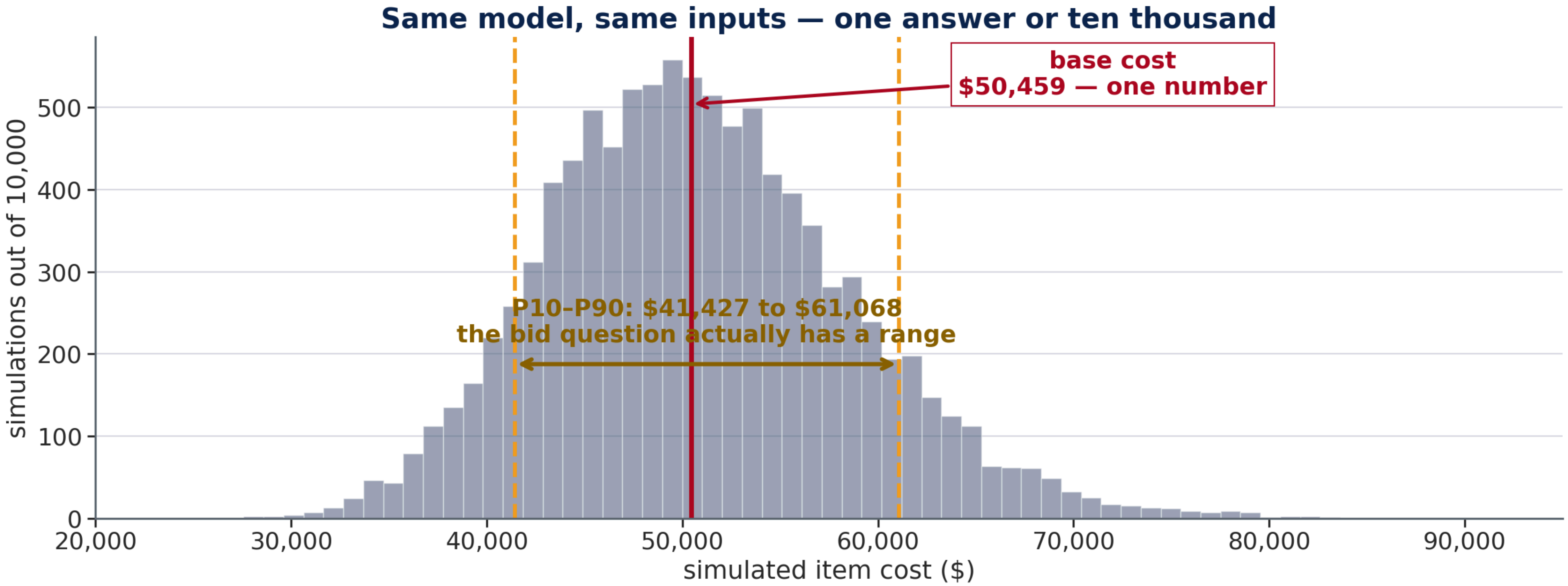
Mean, Median, or Percentile — Which Do You Use?

The simulation hands back an entire distribution. Different decisions read **different parts of it**, and taking the wrong part is a common and expensive mistake.

The decision in front of you	What you read	Here
What will this cost **on average**, over many jobs?	the **mean**	$50,844
What will the **typical** job cost?	the **median**	$50,350
What do I **bid**, to be safe most of the time?	a **high percentile**	P95 = $64,220
How exposed is the number I already wrote?	an **exceedance probability**	P(> $60,000) ≈ 12%

⚠️ An agency programming a hundred jobs a year budgets to the **mean** — over a hundred contracts the highs and lows genuinely do average out. A contractor bidding *this one job* has nothing to average over, and bids to a percentile. Same distribution, two different right answers.

One Number or Ten Thousand



The deterministic estimate is not wrong. It is the middle of a distribution the client never gets to see unless you compute it.

Exceedance Probabilities and the Contingency

Monte Carlo turns model output into a number a decision-maker can act on.

The estimator's question: what contingency do I carry on this item?

Contingency	Bid at	P(overrun)
0%	$50,459	49.3% — a coin flip
15%	$58,027	17.2%
25%	$63,073	6.4%
30%	**$65,596**	**3.7%** ← first below 5%

A contingency **is** a probability statement. That is why Week 12 could not produce one, and why this is the deliverable of the whole four-week block.

Quick Check — What Drives Cost Uncertainty?

60 seconds · rank them

Four inputs, each with its own spread:

Input	Distribution
Quantity	Normal(2,000 CY, σ = 182)
Unit price	$\log_{10}$ Normal, σ = 0.05 — **three quotes in hand**
Overhead	Normal(0.13, σ = 0.0304)
Escalation	Normal(0.05, σ = 0.0182)

Which input owns the largest share of the variance in item cost — and does the answer match Week 12's tornado?

✅ Unit Price — Same Order, Different Proportions

A decomposition and a tornado do not measure the same thing

Input	Week 12 swing	Week 13 variance share
Unit price	**$59,144**	**59.1%**
Quantity	$15,138	36.4%
Overhead	$4,465	3.16%

The **order** is identical. The **proportions** are not: price's swing is 3.9× quantity's, its variance share only 1.6×.

> 💡 Variance squares the swings, so a decomposition compresses ratios a swing ranking spreads out. Neither is wrong — they answer different questions. The tornado says where to spend effort; the decomposition says how much of the total risk each input owns.

Variance Decomposition — How It Is Computed

Run the simulation once per input, with **only that input random** and the other three held at their base values. Because the inputs are independent, the four variances add to the total.

Input	Share of total variance
Unit price	**59.1%**
Quantity	**36.4%**
Overhead	3.16%
Escalation	1.32%

That independence is an assumption, not a fact. Correlated inputs add a covariance term the decomposition has nowhere to put — and the shares stop adding to 100%.

When Inputs Move Together

The decomposition on the last slide assumed the four inputs are independent. It is worth asking whether that is true here.

A hot market raises unit prices — and a hot market is exactly when escalation runs high. Those two inputs are not independent. They share a cause.

For two inputs, the variance of the total is:

$$\mathrm{Var}(X+Y) \;=\; \mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y)$$

Independence sets that last term to zero. **Positive correlation makes it positive**, and the true spread is wider than the simulation reported.

> ⚠️ This is the failure mode that matters most in practice. Sampling correlated inputs independently is **optimistic** — it quietly thins out the case where everything goes wrong at once, which is precisely the case a contingency exists to cover.

Quick Check — Does Correlation Widen It or Narrow It?

60 seconds · commit before you hear the answer

The simulation reported a standard deviation of **$7,716**, drawing all four inputs independently.

You now learn that in this market **quantity and unit price rise together** — busy seasons bring scope growth *and* higher prices. The correlation is about **+0.6**.

Re-run with that correlation in place. What happens to the standard deviation?

- **A)** It falls — the two inputs partly cancel each other
- **B)** It rises — the bad cases now stack
- **C)** Unchanged — correlation moves the mean, not the spread

✅ B — Positive Correlation Stacks the Bad Cases

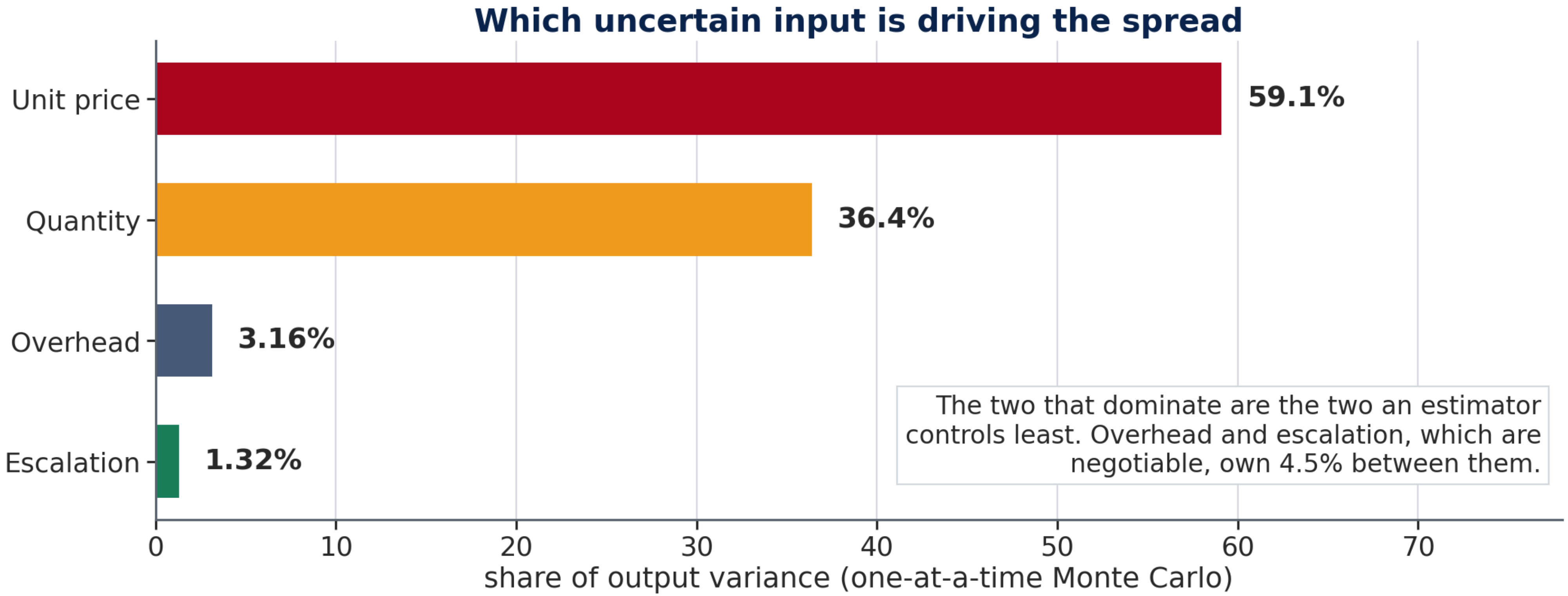
So the contingency set from the independent run was too small

Drawing independently, a high quantity was as likely to meet a low price as a high one, and those mixed cases diluted one another. At **ρ = +0.6** the expensive corner — a lot of work, priced high — comes up far more often.

	Independent	Correlated, ρ = +0.6
Standard deviation	$7,716	**wider**
P95	$64,220	**higher**
Contingency for 5% overrun risk	30%	**more than 30%**

> 💡 Negative correlation does the reverse and narrows the distribution — the reason a diversified programme is safer than one job. Either way the **mean barely moves and the spread does all the changing**, which is how an estimate can look right while the contingency built on it is wrong.

Where the Spread Comes From



Unit price and quantity together account for 95.5% of the variance. Overhead and escalation, which an estimator can actually negotiate, account for 4.5%.

The Number Rests on an Assumption

Every figure in this lecture assumes **three quotes are in hand** — a price spread of about ±12%, $\sigma = 0.05$ in $\log_{10}$ units.

Put the model's own residual spread back — $\sigma = 0.2418$, what the whole market charges — and change nothing else:

	With quotes	Without
Standard deviation	$7,716	**$36,594** (4.74×)
Mean	$50,844	$59,356
Median	$50,350	$50,525
Unit price's variance share	59.1%	**98.2%**

The engineering lesson: the contingency is a statement about the estimator's *information* as much as about the job.

Quick Check — Will the Bid Hold?

90 seconds · estimate the risk

From the simulation: **mean $50,844**, **standard deviation $7,716**.

A project manager proposes bidding this item at **$60,000** — about 19% over the base cost, which sounds generous.

Estimate the probability that the item costs more than $60,000.
(Hint: how many standard deviations above the mean is $60,000?)

- **A)** About 1%
- **B)** About 12%
- **C)** About 25%

✅ B — About 12%, or One Job in Eight

A generous-looking markup is not a safe one

$$z = \frac{60{,}000 - 50{,}844}{7{,}716} = 1.19 \qquad P(Z > 1.19) \approx \mathbf{11.7\%}$$

Bid	Markup	P(overrun)
$50,459	0%	49.3%
$60,000	19%	**12.2%**
$65,596	**30%**	**3.7%** ← the answer

> 💡 The Normal shortcut agrees with the simulation *here*, because this distribution is close to Normal. At the market-wide price spread it is skewed and the z-score is wrong — and the simulation does not care which case you are in.

What Would Actually Reduce the Risk?

The contingency covers the risk. It does not **reduce** it. Three things do, and the decomposition says what each is worth:

Action	What it narrows	Share of variance it touches
More quotes	the price spread	**59.1%**
Commission a take-off survey	the quantity range	**36.4%**
Negotiate a fixed overhead rate	overhead	3.16%

⚠️ The two worth buying are the two that cost money and take time. The one that is free to argue about in a meeting is worth 3%.

Bridge to Machine Learning — Uncertainty Quantification

Monte Carlo concept	ML equivalent	Connection
Input distribution	Feature distribution	How the input varies in real data
N = 10,000 draws	Bootstrapping	Repeated sampling from a data source
Output distribution	Prediction interval	Full uncertainty of a model's outputs
Variance decomposition	Feature importance (SHAP)	Which input drives output variance?
P95 threshold	Safety margin	Risk-calibrated design specification

Monte Carlo is the standard method for uncertainty quantification in ML too. Running a model 1,000 times with random input variations (or with dropout left active) gives a **Monte Carlo estimate of prediction uncertainty** — the same idea applied to a much more complex model.

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Nov 24
In-Class Exercise	The Colab notebook — all sections, plots, and written answers	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

> 💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Nov 24; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 14: Bootstrap Confidence Intervals

Monte Carlo simulates uncertainty by sampling from **assumed distributions** (Normal, etc.). But what if you don't know the distribution?

Bootstrap resamples from the **observed data** itself — no distributional assumption needed.

To estimate the uncertainty in a regression slope:

1. Draw a sample of rows (with replacement) from the dataset
2. Fit the regression on this resample
3. Record the slope
4. Repeat 10,000 times → distribution of slope values → confidence interval

Bootstrap applies to any statistic: slopes, percentiles, correlation coefficients, R^2 values.